



CALCULATION GUIDE FOR THE BOAT DOCKING SUPPORT CALCULATOR

UPC (FNB). BASED ON ISO 12215-5 AND IAP-11
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This document explains the calculation philosophy of the Boat Docking Support Calculator and defines each of the intermediate values shown in the technical detail panel of the application. It is designed for any user who wants to understand what lies behind the results, without needing advanced technical knowledge.

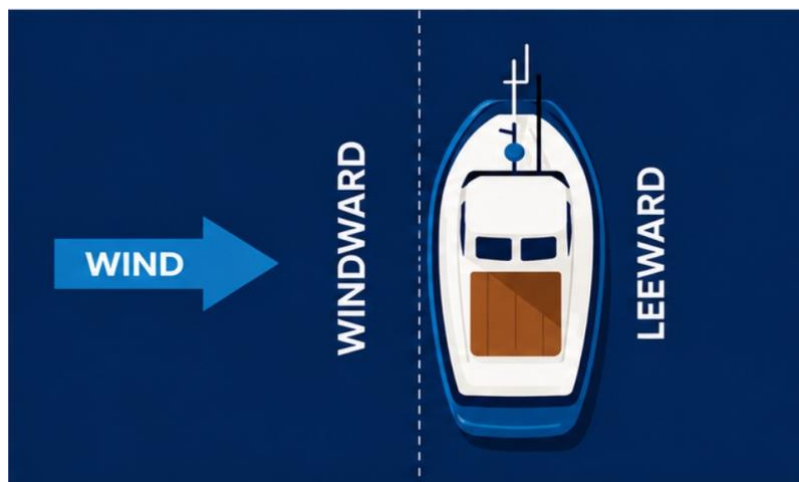
1. Calculation philosophy

The calculator answers a very specific question: how many minimum props are needed to keep a dry-docked vessel safe? The calculation is carried out in two phases:

Phase 1. Self-weight (ISO 12215-5): The pressure the hull must withstand when resting on the props is calculated. From this, the maximum force each prop can bear without damaging the hull (limit force) is obtained. Dividing the vessel's weight by this force gives the minimum number of props without considering wind.

Phase 2. Wind action (IAP-11): Wind pushes the vessel sideways and generates an overturning moment. This means the leeward props must bear an additional load. The calculation determines how many extra props are needed to balance this force and where they must be placed.

The windward and leeward sides are illustrated below to aid understanding of the calculations.



All assumptions are conservative: when there is uncertainty, the model always chooses the value that results in more props. This ensures the tool never recommends an insufficient docking configuration.

2. ISO 12215-5 Variables — Self-weight calculation

These values come from the European standard UNE EN ISO 12215-5, which defines how to calculate the pressure that the hull panels of a recreational vessel must withstand according to its dimensions and type.

mLDC	<i>kg</i>	Estimated mass of the vessel at full load. Since it is generally not known precisely, it is estimated from the waterline length using an empirical formula from the standard. $mLDC = 1369 \cdot LWL - 7223.1$
b = l	<i>m</i>	Reference panel dimension of the hull. It is the side of a square equal to one quarter of the waterline length. It represents the hull area in direct contact with each prop. $b = LWL / 4$
V	<i>knots</i>	Estimated maximum speed of the vessel. For sailboats it is estimated from the waterline length; for motorboats, from the engine power.
nCG	<i>dimen.</i>	Dynamic load factor. Represents the negative acceleration experienced by the vessel when sailing through waves. Always between 3 and 7 per standard.
KL	<i>dimen.</i>	Longitudinal pressure distribution factor. Adjusts pressure according to the panel's position along the hull length. Stern areas receive more pressure than bow areas.
KAR	<i>dimen.</i>	Area reduction factor. Larger panels receive a lower average pressure. The larger the panel, the lower KAR.
P_design	<i>N/m²</i>	Final design pressure adopted. It is the maximum value among all pressures calculated for the hull panels. The most unfavourable value is used to ensure safety.
F_limit	<i>N</i>	Maximum force each prop can bear without damaging the hull. Dividing the vessel's weight by this value gives the minimum number of props. $F_limit = 0.1838 \cdot P_design \cdot b^2$
Δ (Delta)	<i>kg</i>	Hydrostatic displacement: the equivalent weight of the water volume displaced by the vessel. When dry-docked, this weight rests on the props.

3. IAP-11 Variables — Wind action calculation

These values come from the Spanish standard IAP-11, adapted to calculate the horizontal force that wind exerts on the dry-docked vessel and the resulting overturning moment.

v_{b(T)}	<i>m/s</i>	Basic wind speed corrected for the return period (10 years). The reference speed is 29 m/s, corresponding to zone C of the Catalan coast (the most demanding).
c_{prob}	<i>dimen.</i>	Probability coefficient. Adjusts the wind speed according to the chosen return period. For T = 10 years its value is slightly less than 1.
H_{exp}	<i>m</i>	Height of the vessel surface exposed to wind. Estimated as half the chine beam, representing the height of the hull side visible above the ground. $H_{exp} = BC / 2$
A_{ref}	<i>m²</i>	Lateral surface area of the vessel exposed to wind. Calculated as a rectangle of waterline length by exposed height. $A_{ref} = LWL \cdot H_{exp}$
ce(z)	<i>dimen.</i>	Wind exposure coefficient as a function of height. Accounts for the fact that wind is more intense at greater heights.
F_w	<i>N</i>	Total horizontal wind force on the vessel. Pushes the vessel sideways towards the leeward side. $F_w = 0.5 \cdot 1.25 \cdot v_b^2 \cdot ce \cdot 1.65 \cdot A_{ref}$
M_w	<i>N·m</i>	Overturning moment generated by wind. Tends to overturn the vessel towards the leeward side. $M_w = F_w \cdot (H_{exp} / 2)$
2·M_w/B	<i>N</i>	Wind equivalent force expressed as a load on the props. Allows the wind effect to be added to the weight to calculate the total number of props (N_TFM). $2 \cdot M_w / BC$

4. Main results

N_TFG	<i>units</i>	Minimum number of props without wind. Calculated considering only the vessel's self-weight. It is the reference value to quantify the wind impact.
N_TFM	<i>units</i>	Minimum number of props with wind. Incorporates wind action into the calculation. This is the value to be used in practice. Always $\geq$ N_TFG.
ΔN	<i>units</i>	Wind-induced prop increment. Difference between N_TFM and N_TFG. A value of 0 means wind is not the determining factor for this vessel.
d_min	<i>m</i>	Minimum placement distance of props from the vessel's centreline. Placing props at this distance or further guarantees equilibrium against the wind overturning moment.